

Parameter	Value
Forward model	Monte Carlo electron transport
Elastic scattering	screened Rutherford
Stopping power	Joy-Luo modified Bethe
Range	Kanaya-Okayama (1972)
Materials	17, including 6 compounds by weight-averaged Z and A
Free parameter	epsilon, energy per escaping secondary; fitted per material
Fitted epsilon range	57-243 eV
Emission SE1/SE2 split with per-material interaction-volume kernels	
Kernel construction	enclosed-weight CDF from traced trajectories
Kernel fidelity	better than 1% on trajectory quantiles (Fig. S1B)
Detectors	ETD, TLD, BSE
Optional artefacts	specimen charging (off by default)
Validation	backscatter yield within 15% of published, 20% for gold (Fig. 4B)